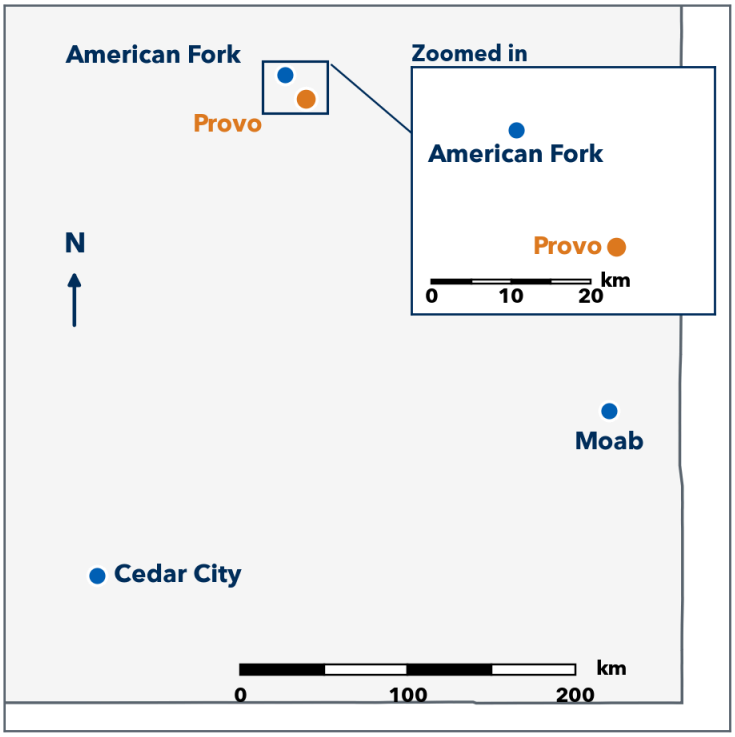
What Is the Temperature in Provo?

CE 414 · Week 9 · in-class worksheet, done on your own

Name: _____ Date and time: _____



1. Gather the samples

Look up the **current air temperature** in each city (weather.com or similar). Find each city's **straight-line distance to Provo** with the scale bar (the zoomed-in box for American Fork) or an online map's measure tool.

City	Temp (°F)	Distance to Provo (km)
American Fork		
Moab		
Cedar City		

Temperature source: _____

2. Your own method

Estimate Provo any way you like from the three temperatures.

Provo: _____ °F How?

3. Thiessen polygons (nearest neighbor)

On the map, join each pair of known cities, draw the **perpendicular bisector** of each line, and join the bisectors into three polygons. Provo takes the value of the polygon it falls in.

Polygon: _____ **Provo:** _____ °F

4. Nearest neighbors, n = 2

Average the temperatures of the **two cities closest** to Provo.

(_____ + _____) ÷ 2 = **Provo:** _____ °F

5. Inverse distance weighted (IDW), power 2

Weight each city by **1 / d²** and take the weighted average of all three.

$$\text{Provo} = \Sigma\,(w \cdot T) / \Sigma\, w, \quad w = 1/d^2$$

City	w = 1 / d²	w · T
American Fork		
Moab		
Cedar City		
Sum		

Provo = Σ(w · T) ÷ Σw = _____ °F

The weights are tiny: keep three significant figures, or use 1 / (d ÷ 10)² instead. Scaling every weight by the same amount does not change the answer.

6. Check

Provo's actual temperature now: _____ °F

Which method came closest, and why?